

Andrey Zakharov

Geospatial Software Engineer

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Geospatial Software Engineer with 3+ years of hands-on project-based development experience. Builds geospatial and simulation software combining Java, Python, OpenStreetMap, DEM/GeoTIFF data, terrain generation and web-based map interfaces. Experienced in designing end-to-end pipelines for geodata ingestion, normalization, spatial processing, procedural generation, external API integration and interactive visualization.

Experience

Independent Project Development: 2023 – Present.

Cartopia – Geospatial Software Engineer

2023 – 2025

Tech: Java 17, Forge, Gradle, Jackson, Gson, Java HttpServer, HTML/CSS/JavaScript, OpenStreetMap, Overpass/Nominatim, DEM/GeoTIFF, OpenLandMap, GMRT/OpenTopography, Open-Meteo

- Built a Java/Forge geospatial generation system that reconstructs real-world areas at approximately 1:1 scale from terrain, roads, buildings, railways, water, vegetation and infrastructure data.
- Designed a multi-source geodata pipeline combining OpenStreetMap, Overpass/Nominatim, DEM/GeoTIFF elevation, land-cover data, weather and time services with terrain smoothing and procedural world generation.
- Developed an extensible architecture with 80+ generator and storage classes for buildings, roads, rails, bridges, tunnels, markings, vegetation, utilities, traffic infrastructure and other city features.
- Implemented a local map-based web UI and HTTP control server for place search, coordinate selection, bounding-box configuration, generation control, progress tracking and player repositioning.
- Added streaming terrain/geodata processing with NDJSON sidecars and terrain-grid storage, plus validation, clipping, retry logic and cleanup for unreliable external data sources.
- Designed the generation workflow for arbitrary worldwide locations selected through address search, coordinates and bounding boxes, with configurable area side lengths starting at 100 m and increasing in 100 m increments without a fixed application-level upper limit.
- Implemented specialized spatial feature handling for crosswalks, parking areas, rail markings, traffic lights, hydrants, EV chargers, power lines, substations, towers, wind turbines, lighthouses and other urban infrastructure.
- Hardened geodata acquisition with bounding-box validation, clipping, connection retries and failure recovery, including GMRT limits of up to 20° per axis and DEM download retries of up to 10 attempts.

UrbanFlow AI – Geospatial Simulation Software Engineer

2026

Tech: Python, FastAPI, OpenStreetMap, SUMO, TraCI, React, TypeScript, Three.js, React Three Fiber, React Leaflet, Pandas, JSON/JSONL

- Built a full-stack geospatial simulation platform that imports and normalizes real OpenStreetMap areas, generates SUMO road networks and scenarios, and visualizes live microscopic traffic in a React/Three.js 3D interface.
- Implemented map search, bounding-box and area-size controls, OSM ingestion, scenario generation and interactive road editing with closures, roadworks, accidents and other spatial events.
- Integrated FastAPI, SUMO and TraCI to manage simulation sessions containing vehicles, pedestrians, buses, trams, public-transport routes and traffic lights.
- Added live map metrics, congestion and access overlays, traffic-light modes, saved simulation artifacts and reproducible experiments with 300–500 vehicles and up to 2,000 controlled traffic lights.
- Designed the OSM-to-SUMO transformation workflow to convert user-selected geographic areas into simulation-ready road networks, junctions, traffic-light systems and public-transport routes.
- Connected React Leaflet map selection with the 3D simulation view, allowing users to define geographic areas, configure scenario parameters and inspect spatial traffic state in real time.
- Made the geospatial simulation pipeline reusable for arbitrary cities and regions covered by OpenStreetMap, with configurable bounding boxes, area sizes, road events and simulation density.

Skills

Technical: Java 17, Python, TypeScript/JavaScript, Minecraft Forge, Gradle, FastAPI, React, Three.js, React Three Fiber, React Leaflet, OpenStreetMap, Overpass, Nominatim, DEM/GeoTIFF, OpenLandMap, GMRT/OpenTopography, Open-Meteo, SUMO/TraCI, Pandas, Jackson, Gson, NDJSON, REST/HTTP APIs, Git.

Languages: Russian – Native; English – C1 (Advanced).

Education

SAINT PETERSBURG STATE UNIVERSITY OF AEROSPACE INSTRUMENTATION (SUAI) – Saint Petersburg, Russia

- Master's Degree, Management – Strategic Human Resource Management 2027
- Bachelor's Degree, Software Engineering – Software Systems Design 2025
- Professional Retraining Diploma – Web and Multimedia Applications Developer 2024
- Additional Professional Education: 250+ certificates and diplomas from 60+ universities, companies, educational platforms, institutes, and professional training programs worldwide, covering diverse academic and professional fields and disciplines.